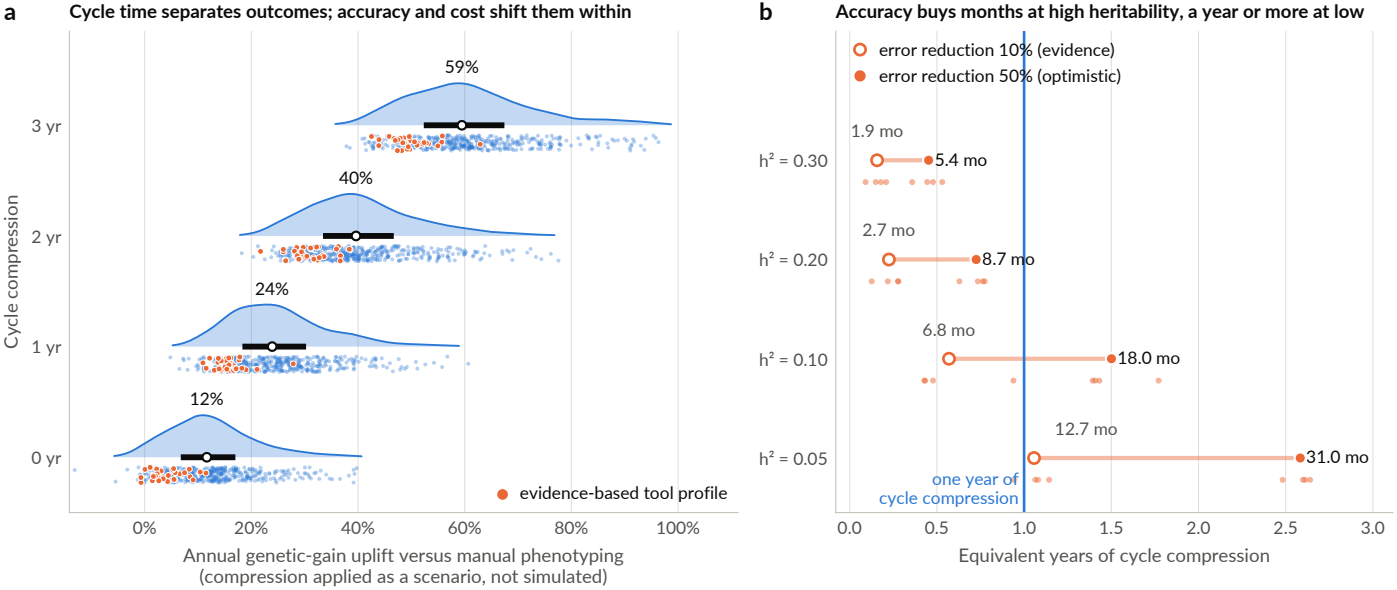

a
Cycle time separates outcomes; accuracy and cost shift them within
Cycle compression
3 yr
2 yr
1 yr
0 yr
59%
40%
24%
12%
evidence-based tool profile
0%
20%
40%
60%
80%
100%
Annual genetic-gain uplift versus manual phenotyping
(compression applied as a scenario, not simulated)
b
Accuracy buys months at high heritability, a year or more at low
error reduction 10% (evidence)
error reduction 50% (optimistic)
h² = 0.30
h² = 0.20
h² = 0.10
h² = 0.05
1.9 mo
5.4 mo
2.7 mo
8.7 mo
6.8 mo
18.0 mo
12.7 mo
31.0 mo
one year of
cycle compression
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Equivalent years of cycle compression